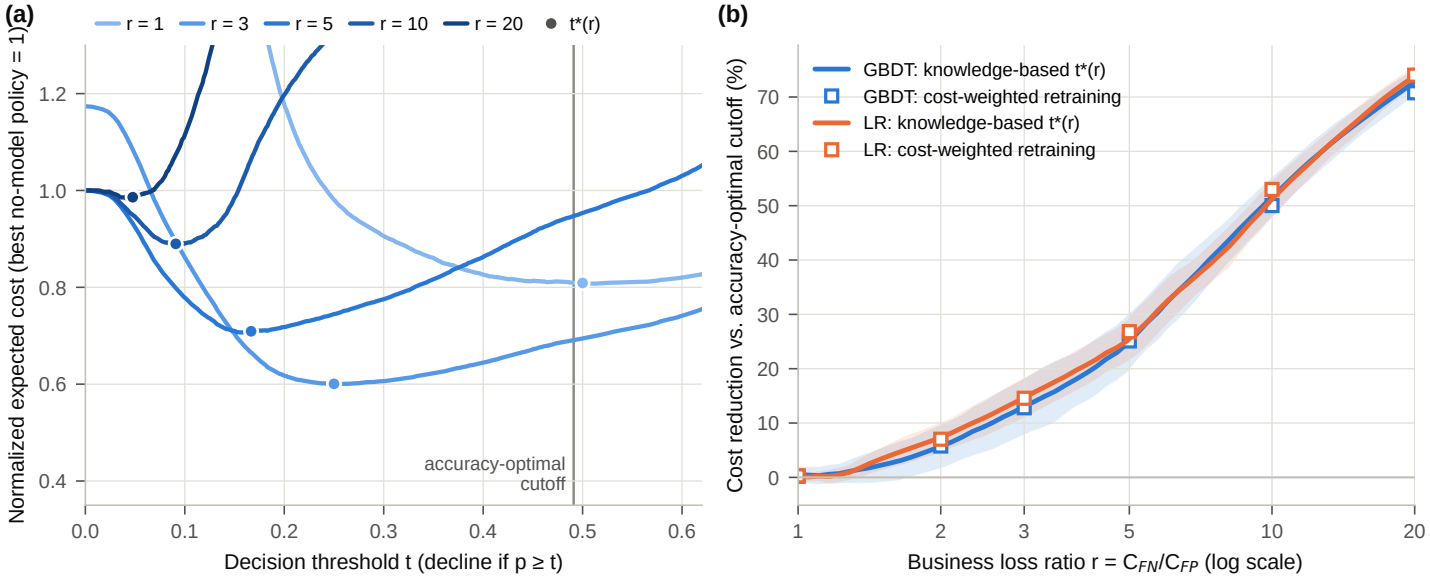
(a)
r = 1
r = 3
r = 5
r = 10
r = 20
t*(r)
Normalized expected cost (best no-model policy = 1)
1.2
1.0
0.8
0.6
0.4
accuracy-optimal
cutoff
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Decision threshold t (decline if p ≥ t)
(b)
GBDT: knowledge-based t*(r)
GBDT: cost-weighted retraining
LR: knowledge-based t*(r)
LR: cost-weighted retraining
Cost reduction vs. accuracy-optimal cutoff (%)
70
60
50
40
30
20
10
0
1
2
3
5
10
20
Business loss ratio r = C_FN/C_FP (log scale)